

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Model Answer Paper Course: B. Tech. Branch: Civil Engineering Semester :V Subject Code & Name: BTHM505 Project Management Max Marks: 60 Date:14/02/2023 Duration: 3 Hr.		
			Marks
Q. 1	Solve Any Two of the following.		12
Q.1.A)	Explain Bar chart, its aspects, advantages and dis-advantages.	(CO1)	6
Ans.	1. A barchart, also known as a bar graph or column chart, is a data visualization tool that uses bars to represent categorical data. 2. The bars are typically arranged vertically, with the height of each bar representing the value of the corresponding category. 3. Barcharts are one of the most common types of data visualization, and they can be used to compare different categories of data, track changes over time, and identify trends. Advantages of using bar charts in construction: 1. Easy to understand: Bar charts are a simple and intuitive way to visualize data, making them easy to understand for everyone involved in a construction project, from project managers to subcontractors to laborers. 2. Effective for comparing data: Bar charts can be used to compare different types of data, such as the cost of different materials, the duration of different tasks, or the productivity of different teams. This can help project managers identify areas where costs can be reduced or efficiency can be improved. 3. Versatile: Bar charts can be used for a variety of purposes in construction, such as planning and scheduling projects, tracking progress, and communicating project status to stakeholders. Disadvantages of using bar charts in construction: 1. Can be complex to create and maintain: Bar charts can become complex, especially for large projects with many tasks and dependencies. This can make it time-consuming and difficult to keep them up to date. 2. Can be difficult to show trends: Bar charts are not well-suited for showing trends over time. For this purpose, line charts or other types of charts are more appropriate. 3. Can be misleading: Bar charts can be misleading if they are not created carefully. For example, if the bars are not scaled consistently, it can be difficult to accurately compare the data.		

		2 marks 2 marks 2 marks	
Q.1.B)	Draw the network and calculate T_E & T_L for all activities for the following Activity 1-2 2-3 2-4 3-5 3-6 4-5 4-7 5-8 6-8 7-8 (i-j) Duration (t^{ij}) Days 5 2 6 4 4 2 3 7 8 2	(CO1)	6
Ans.	Draw Network for given data Calculation for T_E , T_L 	3 marks 3 marks	

Q.1.C)	Prepare table and calculate EST, EFT,LST,LFT, and Total Float associated with the project in Q.no.1(B) , Find critical path & project Duration	(CO1)	6																																																																																							
Ans.	Prepare table and calculate EST, EFT,LST,LFT, Total Float associated with the project Find critical path & project duration Computation of Activity Time and Total Float <table><tr><th rowspan="2">Activity (i—j)</th><th rowspan="2">Duration t^{ij}</th><th colspan="2">Earliest</th><th colspan="2">Latest</th><th rowspan="2">Total Float F_T</th></tr><tr><th>Start time (EST)</th><th>Finish time EFT</th><th>Start time (LST)</th><th>Finish time LFT</th></tr><tr><th>(1)</th><th>(2)</th><th>(3)</th><th>(4)</th><th>(5)</th><th>(6)</th><th>(7)</th></tr><tr><td>1—2</td><td>5</td><td>0</td><td>5</td><td>0</td><td>5</td><td>0</td></tr><tr><td>2—3</td><td>2</td><td>5</td><td>7</td><td>6</td><td>8</td><td>1</td></tr><tr><td>2—4</td><td>6</td><td>5</td><td>11</td><td>5</td><td>11</td><td>0</td></tr><tr><td>3—5</td><td>4</td><td>7</td><td>11</td><td>9</td><td>13</td><td>2</td></tr><tr><td>3—6</td><td>4</td><td>7</td><td>11</td><td>8</td><td>12</td><td>1</td></tr><tr><td>4—5</td><td>2</td><td>11</td><td>13</td><td>11</td><td>13</td><td>0</td></tr><tr><td>4—7</td><td>3</td><td>11</td><td>14</td><td>15</td><td>18</td><td>4</td></tr><tr><td>5—8</td><td>7</td><td>13</td><td>20</td><td>13</td><td>20</td><td>0</td></tr><tr><td>6—8</td><td>8</td><td>11</td><td>19</td><td>12</td><td>20</td><td>1</td></tr><tr><td>7—8</td><td>2</td><td>14</td><td>16</td><td>18</td><td>20</td><td>4</td></tr></table> Answer- Critical Path is (1-2-4-5-8) and project Duration is 20 days FIG. CRITICAL PATH.	Activity (i—j)	Duration t^{ij}	Earliest		Latest		Total Float F_T	Start time (EST)	Finish time EFT	Start time (LST)	Finish time LFT	(1)	(2)	(3)	(4)	(5)	(6)	(7)	1—2	5	0	5	0	5	0	2—3	2	5	7	6	8	1	2—4	6	5	11	5	11	0	3—5	4	7	11	9	13	2	3—6	4	7	11	8	12	1	4—5	2	11	13	11	13	0	4—7	3	11	14	15	18	4	5—8	7	13	20	13	20	0	6—8	8	11	19	12	20	1	7—8	2	14	16	18	20	4	2 marks 2 marks 2 marks
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Q. 2	Solve Any Two of the following.		12
Q.2.A) Ans	Explain briefly Direct cost and indirect cost for construction project • Direct costs are those that can be directly attributed to a specific construction project. • They are typically variable costs, meaning that they fluctuate with the size or scope of the project. Examples of direct costs include: 1. Materials 2. Labor 3. Subcontractors 4. Equipment rental 5. Permits and fees • Indirect costs, on the other hand, are those that cannot be directly attributed to a specific project. • They are typically fixed costs, meaning that they remain relatively constant regardless of the size or scope of the project. Examples of indirect costs include: 1. General and administrative expenses 2. Project management costs 3. Insurance 4. Taxes 5. Depreciation Direct cost: The cost of concrete used to build a foundation is a direct cost, because it can be directly attributed to that specific project. Indirect cost: The salary of the project manager is an indirect cost, because it cannot be directly attributed to any specific project. The project manager oversees multiple projects, and their salary is a fixed cost that does not vary with the size or scope of any individual project. Understanding the difference between direct and indirect costs is important for several reasons. First, it helps to ensure that projects are accurately budgeted. When all costs are properly classified, it is easier to estimate the total cost of a project and track spending against that estimate. Second, understanding direct and indirect costs can help to identify opportunities for cost savings. For example, if a project is over budget, it may be possible to reduce indirect costs by negotiating lower insurance rates or by consolidating administrative functions. Finally, understanding direct and indirect costs is important for complying with government regulations. Many government contracts require contractors to track and report direct and indirect costs separately.	(CO2)	6
		3 marks 3 marks	

Q.2.B)	The following table gives the data for the duration and costs of each activity of project network. The indirect cost of the project 3000Rs/Week.	(CO2)	6																																																							
	<table><thead><tr><th>Activity</th><th>Normal Duration (Week)</th><th>Normal Cost (Rs.)</th><th>Crash Duration (Week)</th><th>Normal Cost (Rs.)</th></tr></thead><tbody><tr><td>1-2</td><td>6</td><td>7000</td><td>3</td><td>14500</td></tr><tr><td>1-3</td><td>8</td><td>4000</td><td>5</td><td>8500</td></tr><tr><td>2-3</td><td>4</td><td>6000</td><td>1</td><td>9000</td></tr><tr><td>2-4</td><td>5</td><td>8000</td><td>3</td><td>15000</td></tr><tr><td>3-4</td><td>5</td><td>5000</td><td>3</td><td>11000</td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></tbody></table>	Activity	Normal Duration (Week)	Normal Cost (Rs.)	Crash Duration (Week)	Normal Cost (Rs.)	1-2	6	7000	3	14500	1-3	8	4000	5	8500	2-3	4	6000	1	9000	2-4	5	8000	3	15000	3-4	5	5000	3	11000																											
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	Draw network. Find project duration and show critical path based on normal duration and corresponding total project cost. Calculate cost slope for each activity.	2 marks 2 marks 2 marks																																																								

Ans.

→ Critical Path

Fig. Network

Solution:

Step 1: Time scaled version of network.

Critical Path: 1-2-3-4 = 15 weeks

Step 2: Cost slopes

Activity	ΔC [₹]	Δt [weeks]	Cost slope [₹/week]
1-2	$14,500 - 7,000 = 7,500$	$6 - 3 = 3$	2,500
1-3	$8,500 - 4,000 = 4,500$	$8 - 5 = 3$	1,500
2-3	$9,000 - 6,000 = 3,000$	$4 - 1 = 3$	1,000
2-4	$15,000 - 8,000 = 7,000$	$5 - 3 = 2$	3,500
3-4	$11,000 - 5,000 = 6,000$	$5 - 3 = 2$	3,000

2.C)	Determine the optimum duration of project and the corresponding minimum cost associated with the project in Q.no.2 (B)	(CO2)	6																												
Ans.	Determine the optimum duration of project – Answer 11 weeks (Min Duration) Determine minimum cost associated with the project- 70,500 Rs (Min. cost) Step 3: Normal duration direct cost. Total normal duration of the project = 15 weeks $\therefore$ Normal duration direct cost = $7000 + 4000 + 6000 + 8000 + 5000$ = ₹ 30,000 Step 4: Activity 2-3 has the least cost slope among the activities. Crashing this activity by 2 weeks, extra cost of crashing = $1000 \times 2 = ₹ 2000$ Project duration after crashing 2-3 by 2 weeks = 13 weeks. Now total direct cost = ₹ 30,000 + ₹ 2,000 = ₹ 32,000 Step 5: Activities 2-3 and 1-3 have least combined cost slope. Hence, crashing these activities by 1 week each, extra cost of crashing = $1000 + 1500 = ₹ 2500$. Project duration after crashing 2-3 and 1-3 by 1 week = 12 weeks. Now total direct cost = ₹ 32,000 + ₹ 2,500 = ₹ 34,500. Step 6: Activity 3-4 is having least cost slope. Hence crashing 3-4 by 1 week, extra cost of crashing = ₹ 3000. Now project duration = 11 weeks. Total direct cost of project = ₹ 34,500 + ₹ 3000 = ₹ 37,500. Step 7: Activities 1-3 and 1-2 are having least combined cost slope. Hence, crashing 1-3 and 1-2 by 2 weeks each, extra cost of crashing = $2 \times 1500 + 2 \times 2500 = ₹ 8000$. Now project duration = 9 weeks Total direct cost of project = ₹ 37,500 + ₹ 8000 = ₹ 45,500. Step 8: Now, all the three paths have become critical. Now crashing 3-4 and 2-4 which are in all the three paths, by one week each, extra cost of crashing = $3000 + 3500 = ₹ 6500$. Now project duration = 8 weeks Total direct cost of project = ₹ 45,500 + 6500 = ₹ 52,000. Since one critical path is fully crashed, further crashing is not possible. Step 9: Total cost of the project <table border="1"> <thead> <tr> <th>Project duration [weeks]</th><th>15</th><th>13</th><th>12</th><th>11</th><th>9</th><th>8</th></tr> </thead> <tbody> <tr> <td>Direct cost [₹]</td><td>30,000</td><td>32,000</td><td>34,500</td><td>37,500</td><td>45,500</td><td>52,000</td></tr> <tr> <td>Indirect cost [₹]</td><td>45,000</td><td>39,000</td><td>36,000</td><td>33,000</td><td>27,000</td><td>24,000</td></tr> <tr> <td>Total cost [₹]</td><td>75,000</td><td>71,000</td><td>70,500</td><td>70,500</td><td>72,500</td><td>76,000</td></tr> </tbody> </table> Hence, the optimum project duration = 11 weeks. The corresponding least cost = ₹ 70500.	Project duration [weeks]	15	13	12	11	9	8	Direct cost [₹]	30,000	32,000	34,500	37,500	45,500	52,000	Indirect cost [₹]	45,000	39,000	36,000	33,000	27,000	24,000	Total cost [₹]	75,000	71,000	70,500	70,500	72,500	76,000	3 Marks 3 Marks	
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Q. 3	Solve Any Two of the following.			12
A)	Differentiate between CPM and PERT		(CO3)	6
Ans.			6 Marks	
	Sr No.	PERT	CPM	
	1.	PERT is that technique of project management which is used to manage uncertain (i.e., time is not known) activities of any project.	CPM is that technique of project management which is used to manage only certain (i.e., time is known) activities of any project.	
	2.	It is event oriented technique which means that network is constructed on the basis of event.	It is activity oriented technique which means that network is constructed on the basis of activities.	
	3.	It is a probability model.	It is a deterministic model.	
	4.	It majorly focuses on time as meeting time target or estimation of percent completion is more important.	It majorly focuses on Time-cost trade off as minimizing cost is more important.	
	5.	It is appropriate for high precision time estimation.	It is appropriate for reasonable time estimation.	
	6.	It has Non-repetitive nature of job.	It has repetitive nature of job.	
	7.	There is no chance of crashing as there is no certainty of time.	There may be crashing because of certain time boundation.	
	8.	It doesn't use any dummy activities.	It uses dummy activities for representing sequence of activities.	
	9.	It is suitable for projects which required research and development.	It is suitable for construction projects	

Q.3B)	Draw the network and find the project duration & Critical path for the following <table><thead><tr><th>Activity (i-j)</th><th>t_o (Optimistic Time)</th><th>t_L (Most likely Time)</th><th>t_p (Pessimistic Time)</th></tr></thead><tbody><tr><td>1-2</td><td>6</td><td>9</td><td>18</td></tr><tr><td>1-3</td><td>5</td><td>8</td><td>17</td></tr><tr><td>2-4</td><td>4</td><td>7</td><td>22</td></tr><tr><td>3-4</td><td>4</td><td>7</td><td>16</td></tr><tr><td>4-5</td><td>4</td><td>10</td><td>22</td></tr><tr><td>2-5</td><td>4</td><td>7</td><td>10</td></tr><tr><td>3-5</td><td>2</td><td>5</td><td>8</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></tbody></table> Solution. Let us first determine the expected time of completion (t_E) and variance σ^2 for each activity using the relations $t_E = \frac{t_o + 4t_L + t_p}{6}$ and $\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$ The computation are arranged in Table <table><thead><tr><th>Activity</th><th>t_o</th><th>t_L</th><th>t_p</th><th>t_E</th><th>σ^2</th></tr></thead><tbody><tr><td>1—2</td><td>6</td><td>9</td><td>18</td><td>10</td><td>4</td></tr><tr><td>1—3</td><td>5</td><td>8</td><td>17</td><td>9</td><td>4</td></tr><tr><td>2—4</td><td>4</td><td>7</td><td>22</td><td>9</td><td>9</td></tr><tr><td>3—4</td><td>4</td><td>7</td><td>16</td><td>8</td><td>4</td></tr><tr><td>4—5</td><td>4</td><td>10</td><td>22</td><td>11</td><td>9</td></tr><tr><td>2—5</td><td>4</td><td>7</td><td>10</td><td>7</td><td>1</td></tr><tr><td>3—5</td><td>2</td><td>5</td><td>8</td><td>5</td><td>1</td></tr></tbody></table> The computation for T_E and T_L and slack for various events are arranged in Table . For the last event, T_L is taken equal to its T_E. From Table , we find that slack is minimum at events 1, 2, 4 and 5. Hence path 1—2—4—5 is the critical path. The critical path is marked by thick lines on Fig.	Activity (i-j)	t_o (Optimistic Time)	t_L (Most likely Time)	t_p (Pessimistic Time)	1-2	6	9	18	1-3	5	8	17	2-4	4	7	22	3-4	4	7	16	4-5	4	10	22	2-5	4	7	10	3-5	2	5	8									Activity	t_o	t_L	t_p	t_E	σ^2	1—2	6	9	18	10	4	1—3	5	8	17	9	4	2—4	4	7	22	9	9	3—4	4	7	16	8	4	4—5	4	10	22	11	9	2—5	4	7	10	7	1	3—5	2	5	8	5	1	(CO3)	6
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Ans.	Draw Network , TE & TL Caluation and Find Critical Path Ans.- Critical pathis is (1-4-5) & Project duration =30days	3 Marks 3 Marks																																																																																									

3.C)	Find Standard Deviation and Z value if the project mentioned in Q. no 3 (B) has the completed in 35 Days	(CO3)	6
Ans	Find Standard Deviation Find Z value Now deviation along the critical path is given by $\sigma = \sqrt{\sum \sigma_y^2}$ where $\sum \sigma_y^2$ = sum of variances along critical path $= \sigma_{1-2}^2 + \sigma_{2-4}^2 + \sigma_{4-5}^2$ $= 4 + 9 + 9 = 22$ $\therefore \sigma = \sqrt{22} = 4.69$ $\therefore Z = \frac{T_s - T_E}{\sigma}$, where $T_s = 35$ days (given) $Z = \frac{35 - 30}{4.69} = 1.066$	3 Marks 3 Marks	

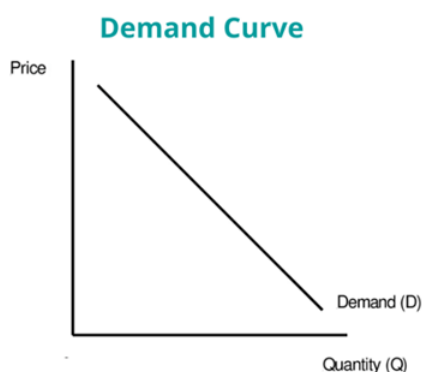
Q.4	Solve Any Two of the following.		12
Q.4.A)	Explain Demand and supply	(CO4)	6
Ans.	Explain Demand Explain supply The law of supply and demand combines two fundamental economic principles describing how changes in the price of a resource, commodity, or product affect its supply and demand. As the price increases, supply rises while demand declines. Conversely, as the price drops supply constricts while demand grows. Levels of supply and demand for varying prices can be plotted on a graph as curves. The intersection of these curves marks the equilibrium, or market-clearing price at which demand equals supply, and represents the process of price discovery in the marketplace. The law of demand holds that the demand level for a product or a resource will decline as its price rises, and rise as the price drops. Conversely, the law of supply says higher prices boost supply of an economic good while lower ones tend to diminish it. A market-clearing price balances supply and demand, and can be graphically represented as the intersection of the supply and demand curves. The degree to which changes in price translate into changes in demand and supply is known as the product's price elasticity. Demand for basic necessities is relatively inelastic, meaning it is less responsive to changes in their price. The Law of Demand The law of demand holds that demand for a product changes inversely to its price, all else being equal. In other words, the higher the price, the lower the level of demand. Because buyers have finite resources, their spending on a given product or commodity is limited as well, so higher prices reduce the quantity demanded. Conversely, demand rises as the product becomes more affordable. As a result, demand curves slope downward from left to right, as in the chart below. Changes in demand levels as a function of a product's price relative to buyers' income or resources are known as the income effect.	3 Marks 3 Marks	

Demand and supply analysis is the study of price changes with respect to the number of buyers and sellers for a particular product.

In a state of market equilibrium, the demand curve and the supply curve intersect. At this point, the demand and supply for a commodity are equal. The fair price of farm yield is also based on demand-supply. In a perfectly competitive market, farmers are price takers and demand-supply is the price maker. Therefore, governments set a minimum price to protect farmers.

The demand for goods and services signifies the consumers' need, willingness, and ability to buy a particular product. Supply, on the other hand, refers to the production capacity of manufacturers and distributors. A demand and supply analysis is essential to understand the impact of those forces on buyers, sellers, buyer-seller interactions, and prices.

The demand-supply study is based on the laws of demand and supply. The law of demand depicts an inverse relationship between goods price and goods demand. The law of demand renders a downward sloping curve—demand goes up when goods price falls.



If other factors remain constant, the law of supply suggests that an increase in goods price should result in an increase in the supply of goods and vice-versa. It establishes a direct relationship between the price and supply of a commodity forming an upward curve.



Q.4B)	Explain different types of interests	(CO4)	6
Ans.	Types of interests (Explain any 4 types) Marks $1.5 \times 4 = 6$ marks Fixed interest Variable interest Annual percentage Rate Simple interest Compound interest 1. Fixed interest type A fixed interest type remains the same throughout the fixed rate term, often 2-5 years. An example of a fixed interest type would be a mortgage. With a mortgage, you (the borrower) agree to make regular payments to the lender over a set period of time, known as the mortgage term. The interest rate is “fixed” for a set period, meaning it will not go up or down. Take a 30-year interest only mortgage for £100,000 at an interest rate of four percent. This means your annual interest would be £4,000. 2. Variable interest type Whereas the prior type of interest is a set rate, so you know exactly what you’re paying, a variable interest type, as the name suggests, changes over time. For example, if the agreed interest rate is 3.95%, a bank could set its variable interest rate at 3.30%. However, the next month, the base rate could change to 4.1%, and the variable rate could rise to 3.45%. An example of this could be taking out a loan for £10,000 with a variable interest rate of 3%. If you take out a loan for £5,000 over five years at a variable interest rate of three percent, your monthly repayments would be £93.22, and you would pay back a total of £5,593.20. This includes the £5,000 you borrowed, plus £593.20 in interest. If the interest rate changes to five percent, your monthly repayments would increase to £96.30, and you would pay back a total of £5,775.60. This includes the £5000 you borrowed, plus £775.60 in interest. 3. Annual Percentage Rate An annual percentage rate (or APR) is the total cost of a loan, including both the interest rate and any other associated fees. This is usually displayed as a percentage and will be charged annually. For example, if you take out a loan for £5,000 with an APR of 20%, you will be charged £1,000 in total interest and fees over the course of the year.	$1.5 \times 4 =$ 6marks	

	APR is a standardised measure that is used to make it easy for borrowers to compare the cost of borrowing easy, even where interest and fees vary wildly. 4. Simple interest type Simple interest is a type of interest that's only charged on the principal or the initial amount of money borrowed. It's not compounded, which means that interest is not charged on interest that has already been applied to the loan. For example, if you borrow £100 at a simple interest rate of five percent, you will be charged £5 in interest over the course of a year. This includes the £100 you borrowed, plus £5 in interest. Simple interest is often used for short-term loans, such as a personal loan or car loan as well as mortgages. 5. Compound interest type Compound interest is a type of interest that's charged on the principal amount, as well as any accumulated interest. This means that the interest is added to the principal amount and then charged again. For example, let's say you took out a loan for £100 with a 5% interest rate, your payments would look a little like this; Month 1 – (Starting Balance) £100 + (Interest) = £105 You make a payment of £10, so you're remaining balance to pay is £95 Month 2 – (Starting Balance) £95 + (Interest) = £99.75 You make a payment of £10, so your remaining balance to pay is £89.75 Month 3 – (Starting Balance) £89.75 + (Interest) = £94.23		
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Q.4C)	Explain the importance of economics in the construction industry(Civil Engineering field).	(CO4)	6
Ans.	Importance of economics in the construction industry(Any six points) One major sector of the economy which is receiving increasing attention as a possible area for the development of better institutional structure in the emerging nations is construction, an industry present in every developmental activity. Construction contributes to economic development by satisfying some of the basic objectives of development including output generation, employment creation, and income generation, and re-distribution. It also plays a major role in satisfying basic physical and social needs, including the production of shelter, infrastructure, and consumer goods. A major objective of development is economic growth. During the early developmental process economic growth seems to be generally high, and construction, along with manufacturing, tends to play an increasingly important role in the economy while agriculture's importance declines; once a relatively high level of development is achieved, however, economic growth appears to slow, and construction's role tends to stabilize or even decline slightly while the other sectors continue as before. The Construction Industry is an investment-led sector where the government shows high interest. While construction's direct contributions to development are significant, it also stimulates a sizeable amount of economic growth through backward and forward linkages. Construction's requirements for goods and services from other industries are considerable; the development of the construction industry therefore stimulates these ancillary industries, thus encouraging further economic growth. In developing countries, the construction of physical facilities makes up more than one half of gross domestic investment and tends to be concentrated on basic infrastructure in agriculture, mining, transportation, communication, and utilities. Infrastructural services make some contribution to GDP, but they also stimulate the development of other industries which, in turn, contribute more directly to economic growth. Construction plays a pivotal role in the growth of the economy as a basic infrastructure facility. The health of the economy is often gauged by monitoring new home construction starts as an indicator of consumer confidence. An increase in the level of construction activity will increase the GDP and have a positive influence on employment, and public income.	6marks	

	Any budgetary squeeze on construction will reduce the construction activity and after a time lag, reflect with decreased GDP and slowing down the economic growth. Any change in the level of construction output will have a great corresponding effect on the country's economy as a whole. Throughout the world, construction is a leading industry in all market economies. It is the first indicator of the health of the economy. Its acceleration starts a buoyant economic growth and vice-versa. Korea, Taiwan, and Hong Kong have used the construction sector to lift themselves into vibrant economics. Malaysia and China are using the same strategy, by rebuilding their cities and highways, etc. to become major players in the global economy. According to the U.S. Census, spending on construction projects totaled more than \$993 billion in 2014. Nowadays, throughout the world, construction has become a leading industry in all market economies. The construction industry accounts for more than 10% of global GDP (6-9% in developed countries) and employs around 7% of the global workforce (around 273m+ people). The output of the global construction industry was worth an estimated \$10.8 trillion in 2017.		
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Q. 5	Solve Any Two of the following.		12
Q.5A)	Explain the uses, advantages, and limitations of Break-even analysis.	(CO5)	6
Ans.	Introduction and uses of Break-even analysis. Advantages of Break-even analysis. Limitations of Break-even analysis. A break-even analysis is a financial calculation that weighs the costs of a new business, service or product against the unit sell price to determine the point at which you will break even. In other words, it reveals the point at which you will have sold enough units to cover all of your costs. At that point, you will have neither lost money nor made a profit. A break-even analysis is a financial calculation used to determine a company's break-even point (BEP). It is an internal management tool, not a computation that is normally shared with outsiders such as investors or regulators. However, financial institutions may ask for it as part of your financial projections on a bank loan application. The formula takes into account both fixed and variable costs relative to unit price and profit. Fixed costs are those that remain the same no matter how much product or service is sold. Examples of fixed costs include facility rent or mortgage, equipment costs, salaries, interest paid on capital, property taxes and insurance premiums. Variable costs rise and fall according to changes in sales. Examples of variable costs include direct hourly labor payroll costs, sales commissions and costs for raw material, utilities and shipping. Variable costs are the sum of the labor and material costs it takes to produce one unit of your product. Total variable cost is calculated by multiplying the cost to produce one unit by the number of units you produced. For example, if it costs \$10 to produce one unit and you made 30 of them, then the total variable cost would be $10 \times 30 = \\$300$.	2 marks 2 marks 2 marks	

Advantages of Break-even analysis

Pricing

Break-even analysis is a very valuable technique for a corporation, and it has a lot of benefits. It demonstrates how many things they must sell in order to make a profit. It determines if a product is worth selling or is too dangerous to sell. It indicates how much money the company will make at each level of output.

Gaining funds

When it comes to collecting financing, break-even analysis is usually an important part of a company's strategy. If you want to get funding for your firm or startup, you'll almost certainly need to do a break-even study. Furthermore, a low break-even point will likely help you feel more at ease about taking on extra debt or funding.

Setting revenue targets

A break-even analysis may also be a useful tool for determining precise sales goals for your team. When you have a precise quantity and a timeframe in mind, it's typically easier to decide on revenue goals.

Reduce risk

Some company concepts are simply not intended to be followed. Break-even analysis can help you reduce risk by guiding you away from investments or product lines that are unlikely to be successful.

Relying on accurate data

Costs can sometimes be classified as both fixed and variable. This can make computations difficult, and you'll almost certainly have to fit them into one of the two.

Competitors are ignored

As a newcomer to the market, you will have an impact on rivals and vice versa. They might modify their pricing, affecting demand for your goods and forcing you to adjust your prices as well. If they expand swiftly and a raw resource that you both use becomes scarce, the price may rise.

Pays of fixed expenses

Most people think about price in terms of how much it costs to make their product. These are referred to as variable costs. You must still pay for fixed expenditures like insurance and web development. You may achieve this by performing a break-even analysis.

Make better choices

Entrepreneurs frequently make decisions based on their emotions. If they are enthusiastic about a new enterprise, they will pursue it. It's necessary to know how you feel, but it's not enough.

Entrepreneurs that are successful make judgments based on facts. When you've put in the effort and have meaningful data in front of you, making a decision will be much easier.

Limitations of Break-even analysis

The assumption behind break-even analysis is that all costs and spending can be clearly divided into fixed and variable components. In reality, however, a clear distinction between fixed and variable expenses may be difficult to make.

Assuming that the selling price remains constant results in a straight revenue line, which may or may not be accurate. The selling price of a product is determined by a variety of factors such as market demand and supply, competition, and so on, and it seldom remains constant.

In actuality, it's rare to discover the assumption that just one product will be created or that the product mix would remain stable.

It presupposes that production and sales quantities will be equal, and that there would be no change in the opening and closing stock of completed goods; however, this is not true in actuality.

The quantity of capital used in the firm is not taken into account in the break-even analysis. In reality, the amount of capital utilized is a key factor in determining a company's profitability.

It is proven to be inappropriate in sectors such as shipbuilding. If fixed expenditures are not taken into account while valuing work in progress, losses may occur each year until the contract is finished. It may result in income tax issues.

A corporation may choose to place an excessive order at a cheaper price based on the marginal cost concept, ignoring plant capacity. It may entail extra labor and the expansion of manufacturing capacity, both of which might raise production costs and cause changes in fixed expenses. Often, the company will lose money.

Fixed costs are assumed to be constant at all levels of activity. Fixed expenses, it should be mentioned, tend to vary after a given degree of activity.

It is assumed that variable costs are proportional to output volume. They move in correlation with production volume in practice, although not always in exact proportions.

Sales income and variable expenses do not grow in lockstep with the production value. They are less proportional than they should be at greater levels of output. This is due to trade discounts, bulk buying economies, concessions for bigger sales, and so on. (source)

The distribution of fixed costs across a number of items is problematic, and it believes that business circumstances will remain constant, which is not the case.

Q.5B)	Explain the importance of Total Quality Management (TQM).	(CO5)	6
Ans.	Importance of Total Quality Management (TQM). (Any six points) The word TQM comprises three terms Total, Quality and Management. Let us expand each term separately, Total: Total means including all. Quality: Quality refers to excellence. Management: Management is the art of maintaining and completing the task. In simple words, total quality management or TQM is defined as a management approach to deliver long term product success and success satisfaction. In TQM all members of the organization work together to improve the processes, products, services and the culture of their workplace. Thus, in scientific terms Total Quality Management is defined as a customer-oriented process. Total quality management aims at the regular enhancement of the business. TQM ensures that all components of the industry work together with a united goal to improve the quality of its services and products. Total quality management usually focuses on enhancing the production process and process of rendering of services. The whole process focuses on the performance to govern the progress of the organization. Importance of Total Quality Management As the name total quality management itself suggests that TQM is essential for implementing quality products and processes within the organization. Here are the other functions of TQM except quality delivery. It establishes the product quality specifications and detailed specification of products. It makes the written obligations and necessities of an intermediate process available for the employees, easing the entire process for them. The principles of TQM help in characterizing the material quality designs and approval guidelines. TQM also helps in infrastructure design and construction. It also helps in the selection of efficient equipment. TQM also provides proper training to the employees ensuring them to discharge their duty in a proper manner. TQM principles also control the shipping and handling of products using inventory control systems. Quality assurance and quality control are the prime functions of total quality management.	6 marks	

	TQM allows the organization to approve the services only after assessing and being assured that the entire process has been completed in a satisfactory manner. TQM balances a healthy environment between the employees and the organization. TQM serves the sensitive needs of customers and makes the industries respond to the complaints and needs of the customers.		
Q.5C)	Explain the uses of computer software (Microsoft Project and Primavera) in project management.	(CO5)	6
Ans.	Uses of Microsoft Project software Microsoft Project Features Microsoft Project, a renowned project management software, is packed with a host of features that streamline project execution: • Task Management: Microsoft Project makes it effortless to assign, track, and manage project tasks for Project Managers. Organizing tasks, setting priorities, and defining milestones, ensures that teams remain focused and projects stay on track. • Resource Management: Efficient use of resources is critical for project success. Microsoft Project assists in optimally allocating and monitoring resources, whether they are human, financial, or material. It helps in avoiding resource over-allocation and enables adjustments to ensure resource availability aligns with project timelines. • Gantt Charts: A picture is worth a thousand words, and this holds true for project schedules. Gantt charts in Microsoft Project provide a visual representation of task sequences, durations, and dependencies, enhancing understanding and communication of project timelines among team members and stakeholders. • Time Tracking: Time is of the essence in project management. Microsoft Project allows for close monitoring of the time spent on various project tasks, which is essential for managing deadlines and assessing team productivity. It helps in ensuring that tasks are completed within the expected timeframes. • Collaboration Tools: Teamwork makes the dream work, and collaboration is key. Microsoft Project's collaboration tools facilitate seamless	3marks 3marks	

	communication and information sharing among team members. Through features like shared documents and discussion boards, it fosters a collaborative environment that enhances team synergy. • Reporting: To make informed decisions, project managers need data. Microsoft Project enables the creation of customizable reports that offer insights into various project aspects. These reports are crucial for evaluating project health and making necessary adjustments.		
	Uses of Primavera software • Project planning & scheduling: Primavera P6 offers Gantt charts, project tables and network diagrams as its main project planning and scheduling tools. These tools allow users to create scope, schedule and resource baselines. • Project portfolio management: It can manage multiple projects in a program or portfolio at once with the enterprise project structure (EPS) tool. • Resource management: Keep track and reallocate resources as needed by using customizable resource leveling forms. • Risk management: Risk analysis features allow project managers to identify, track and resolve risks before they become issues. • Contract management: Manage multiple projects, and get info from the database fast. • Project Reporting: Report on timelines, resources and costs.		